

DATA VISUALIZATION INTERPRETATION

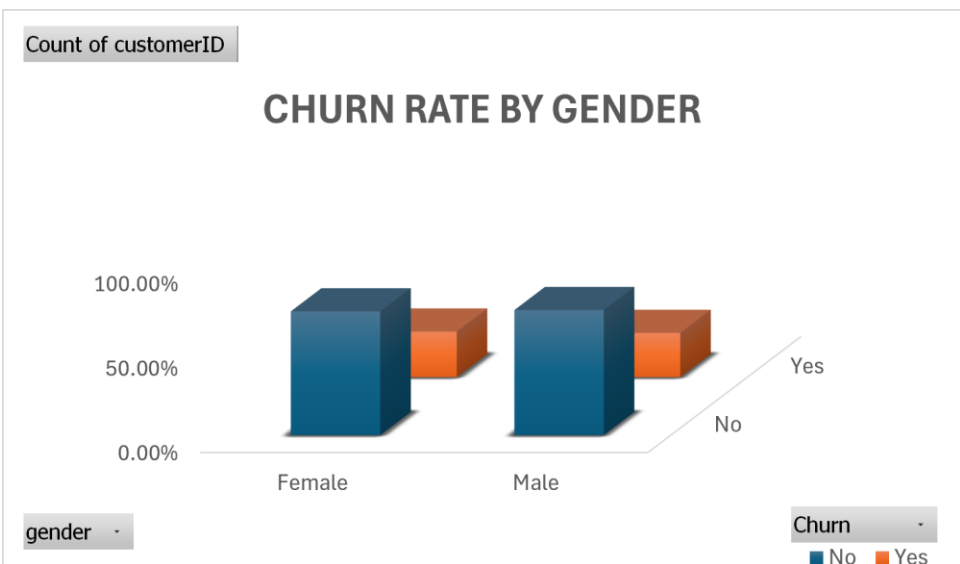
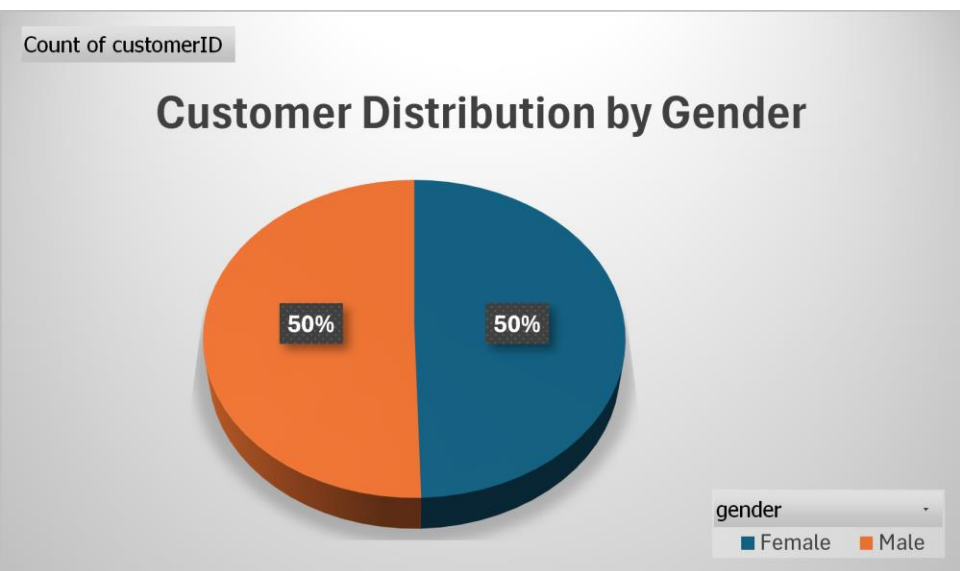
Customer Churn Analysis

Interpreting the visualizations behind customer churn and retention

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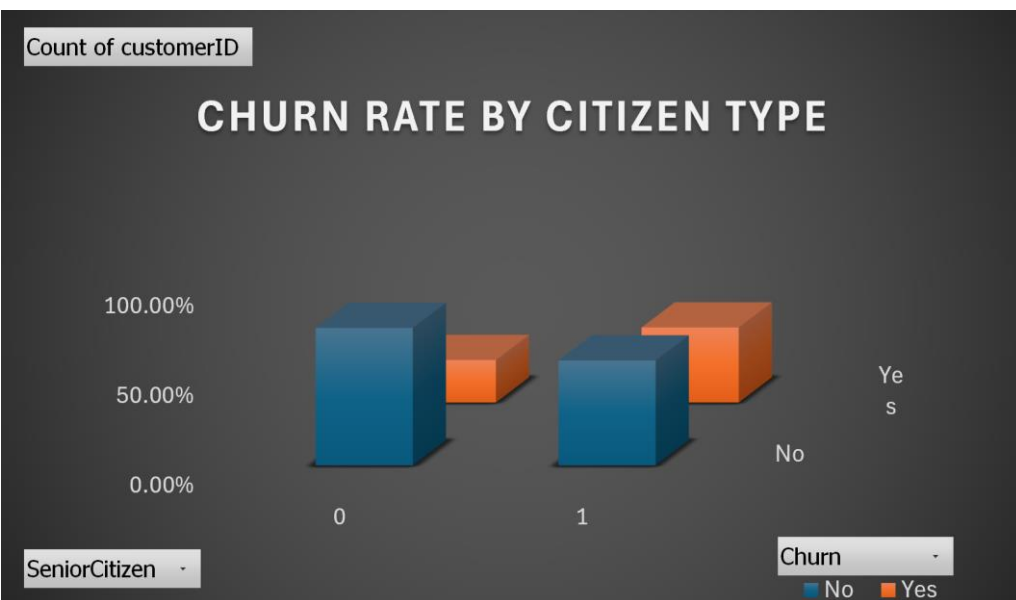
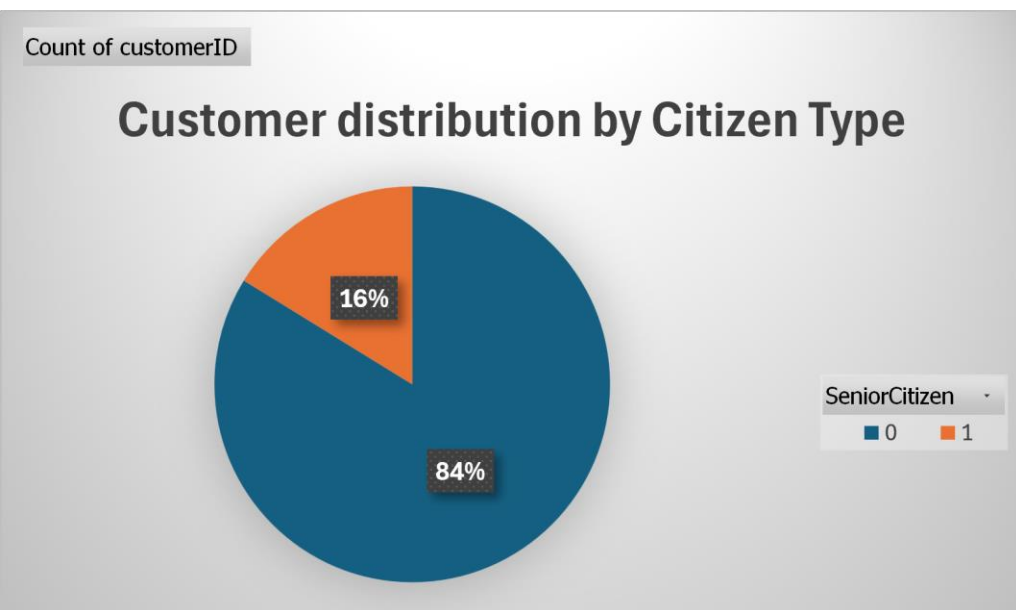


Key Finding

- The customer base is evenly distributed by gender, with approximately 50% female and 50% male customers.
- The churn rates for both genders are also very similar, with only a slight difference between male and female customers.

Business Interpretation: Although the company serves male and female customers in almost equal proportions, gender does not appear to be a significant driver of customer churn. Both groups exhibit similar retention and churn behavior, suggesting that customers leave for reasons unrelated to gender.

Relationship Between the Two Charts: The first chart shows that the customer population is balanced by gender, while the second chart demonstrates that this balance is reflected in nearly identical churn rates. Together, these visuals indicate that gender has little influence on customer retention, suggesting that the company should focus its retention efforts on other factors such as contract type, internet service, payment method, billing preference, and tenure, which showed much stronger relationships with churn in the analysis.

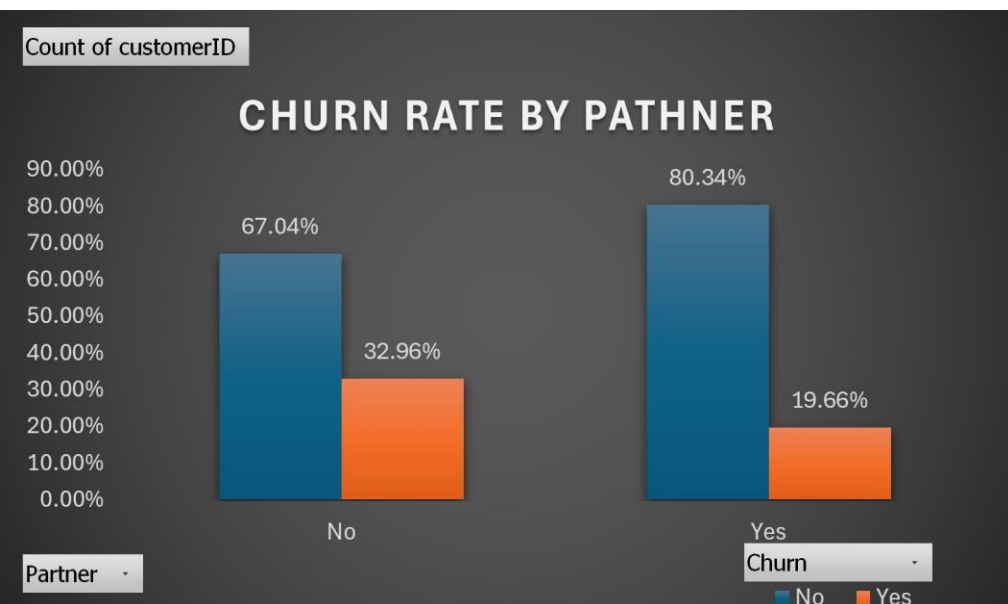
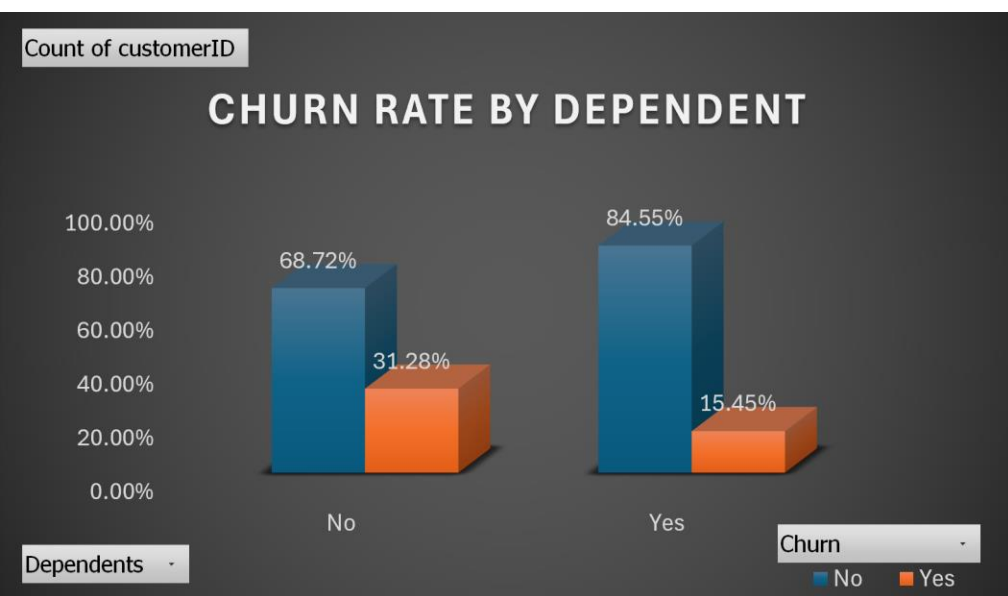


Key Finding

- The customer base is made up of approximately 84% non-senior citizens and 16% senior citizens.
- Despite representing a much smaller portion of the customer base, senior citizens have a noticeably higher churn rate than non-senior customers.

Business Interpretation: Senior citizens are more likely to discontinue their services than non-senior customers. This indicates that age is an important factor influencing customer churn, and senior customers may face challenges such as affordability, service complexity, or unmet service expectations that increase their likelihood of leaving.

Relationship Between the Two Charts: The first chart shows that most customers are non-senior citizens, while the second chart reveals that senior citizens churn at a significantly higher rate despite making up only a small percentage of the customer base. Together, these charts suggest that senior citizens represent a high-risk customer segment. Although they account for only about 16% of customers, they are disproportionately more likely to churn, making them an important group for targeted retention strategies.



Key Finding

- Customers without dependents recorded a churn rate of 31.28%, compared to 15.45% for customers with dependents.
- Similarly, customers without a partner had a churn rate of 32%, while customers with a partner had a lower churn rate of 19%.

Business Interpretation: Customers with stronger family commitments, such as having a partner or dependents, are more likely to remain with the company. In contrast, customers without these commitments are significantly more likely to churn, suggesting that household stability is associated with higher customer retention.

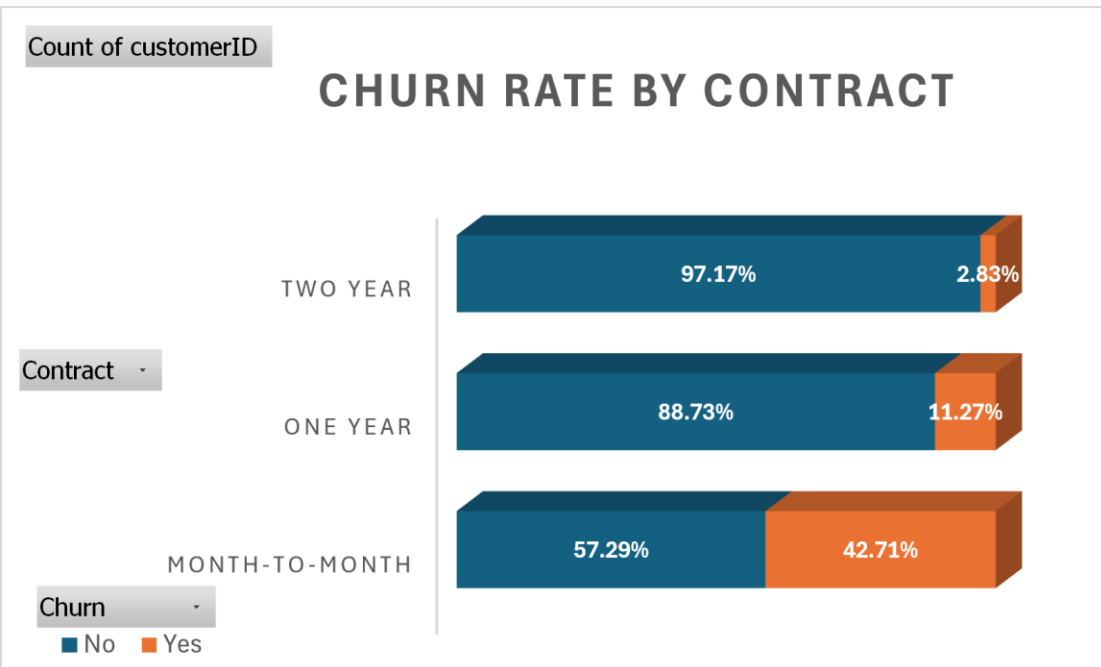
Relationship Between the Two Charts: The relationship analysis shows that customers with partners and dependents are considerably less likely to churn. This suggests that family-oriented customers tend to maintain longer relationships with the company, while customers without partners or dependents represent a higher-risk segment that may require targeted retention efforts.

Key Finding

- Customers on Month-to-Month contracts have the highest churn rate (42.71%), while customers on One-Year (11.27%) and Two-Year (2.83%) contracts are significantly more likely to remain with the company.

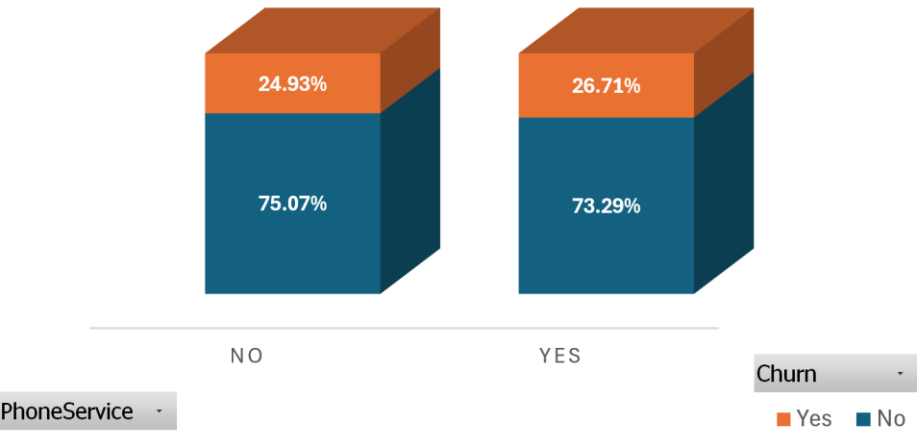
Business Interpretation: The analysis shows a strong relationship between contract type and customer retention. Customers with shorter, month-to-month contracts can leave the service at any time, making them much more likely to churn. In contrast, customers who commit to longer-term contracts demonstrate greater loyalty and are far less likely to cancel their subscriptions.

Business Implication: Contract duration is a major driver of customer retention. Encouraging customers to move from month-to-month plans to one-year or two-year contracts could significantly reduce churn, improve customer lifetime value, and provide the company with more predictable recurring revenue.



Count of customerID

CHURN RATE BY PHONE SERVICES



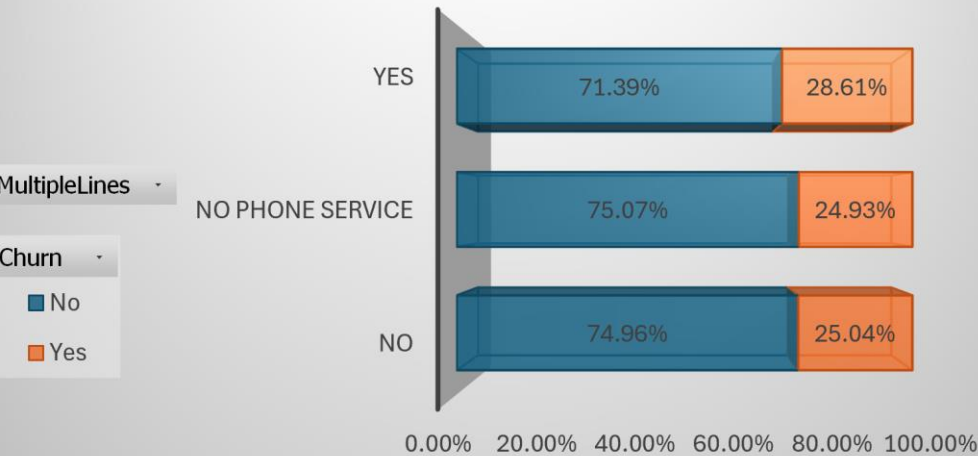
Key Finding

- Customers with and without phone service have almost identical churn rates (26.71% vs. 24.93%), indicating that simply having a phone service has little impact on customer churn.
- Customers with multiple phone lines show a slightly higher churn rate (28.61%) than customers with a single line (25.04%) or no phone service (24.93%). However, the difference is relatively small.

Business Interpretation: Phone service alone does not appear to be a major factor influencing customer retention. Likewise, having multiple phone lines is associated with only a slight increase in churn, suggesting that voice service features have limited influence on customers’ decisions to leave.

Count of customerID

Churn Rate by Multiple lines

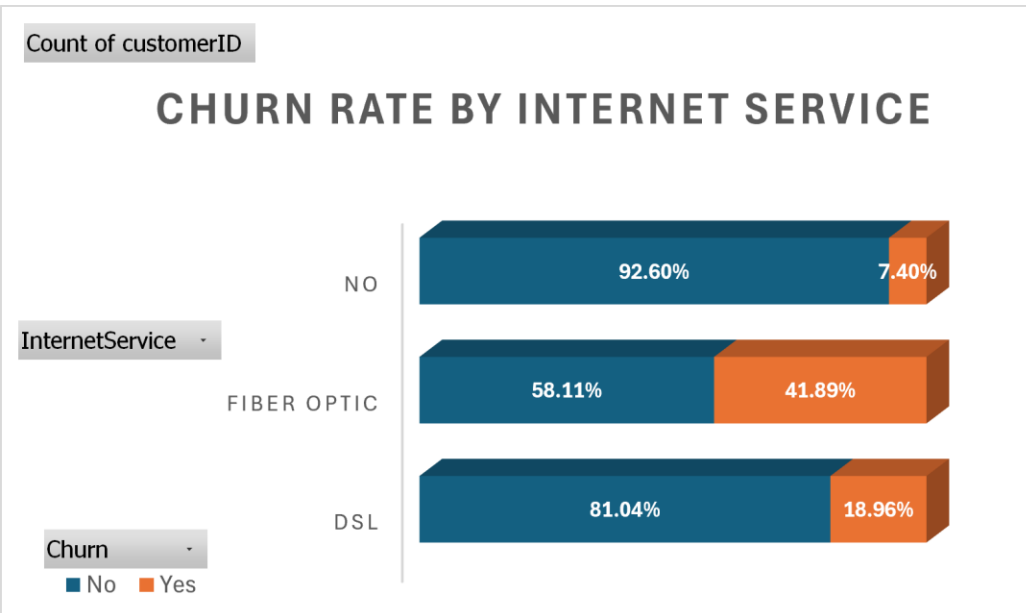


Relationship Between the Charts: The first chart shows that phone service ownership has little effect on churn, while the second chart further explores phone service customers by separating them into single-line and multiple-line users. Although customers with multiple lines exhibit a slightly higher churn rate, the increase is modest. Together, both charts indicate that phone-related services are not primary drivers of customer churn and that other factors are likely to have a stronger influence on customer retention.

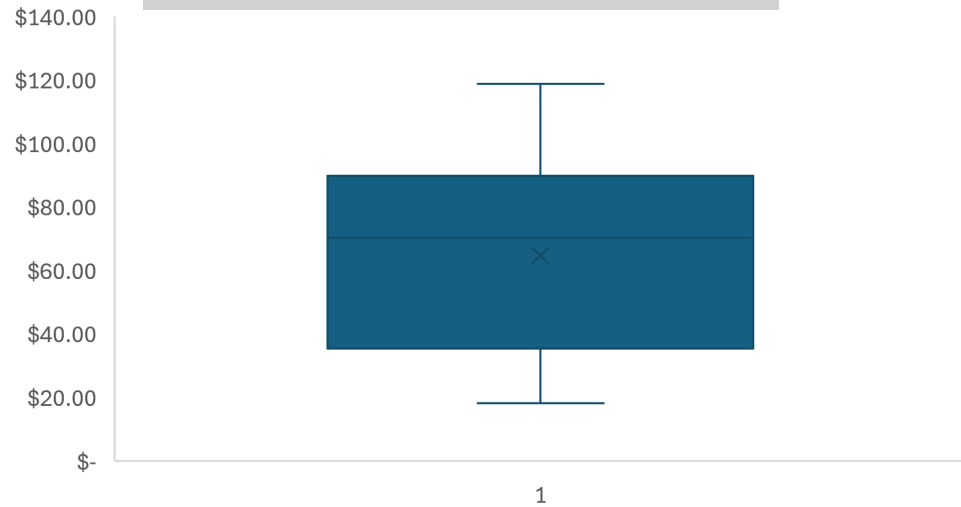
Interpretation

- Customers using Fiber Optic internet service recorded the highest churn rate at 41.89%, which is significantly higher than customers using DSL (18.96%) or those with no internet service (7.40%).
- Customers without internet service showed the highest retention rate (92.60%), indicating they are the least likely to leave.

Business Implication: The high churn among Fiber Optic customers suggests that this service may not be meeting customer expectations. This could be due to factors such as pricing, service quality, network reliability, or customer support. Since Fiber Optic customers contribute the largest proportion of churn, improving their overall experience should be a priority, as it is likely to have the greatest impact on reducing customer churn.



CHARGE DISTRIBUTION



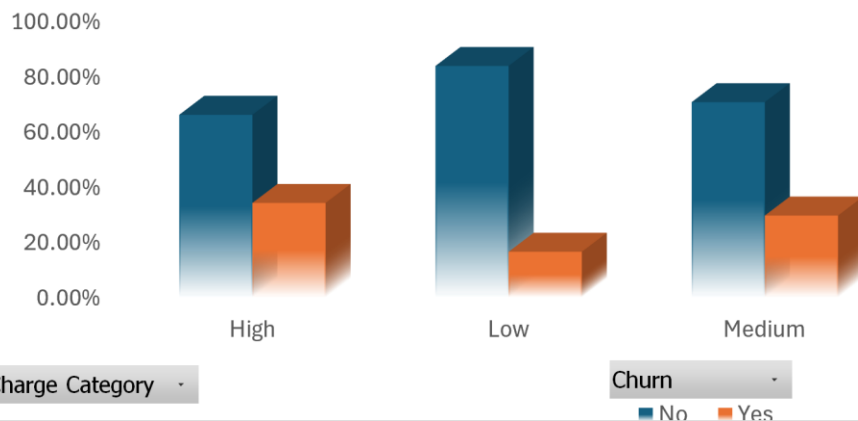
Key Finding

- The box plot shows that Monthly Charges are fairly evenly distributed, with a median monthly charge of \$70.35 and no significant outliers, indicating that customers generally pay within a similar price range.
- The churn chart shows that customers in the High Monthly Charge category experience the highest churn rate compared to customers in the Medium and Low charge categories.

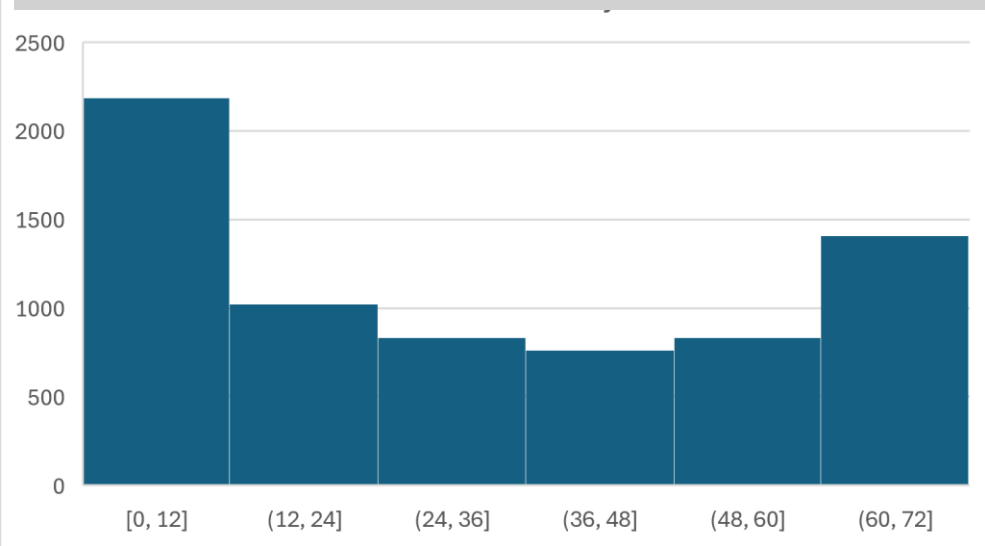
Business Interpretation: Although monthly charges are distributed fairly evenly across customers, churn is more common among customers paying higher monthly charges. This suggests that the amount customers pay each month may influence churn, particularly for customers in the high-charge category.

Relationship Between the Charts: The box plot provides context by showing that monthly charges are evenly spread across the customer base without extreme values. The churn-by-charge chart then demonstrates that, despite this balanced distribution, customers in the High charge category are more likely to churn than those in the Medium or Low categories. Together, these charts indicate that the higher churn among high-paying customers is not simply because a few customers pay unusually high prices. Instead, it suggests that customers paying higher monthly charges may perceive less value for the price they pay or may have higher service expectations that are not being met.

CHURN RATE BY CHARGE



CUSTOMER DISTRIBUTION BY TENURE

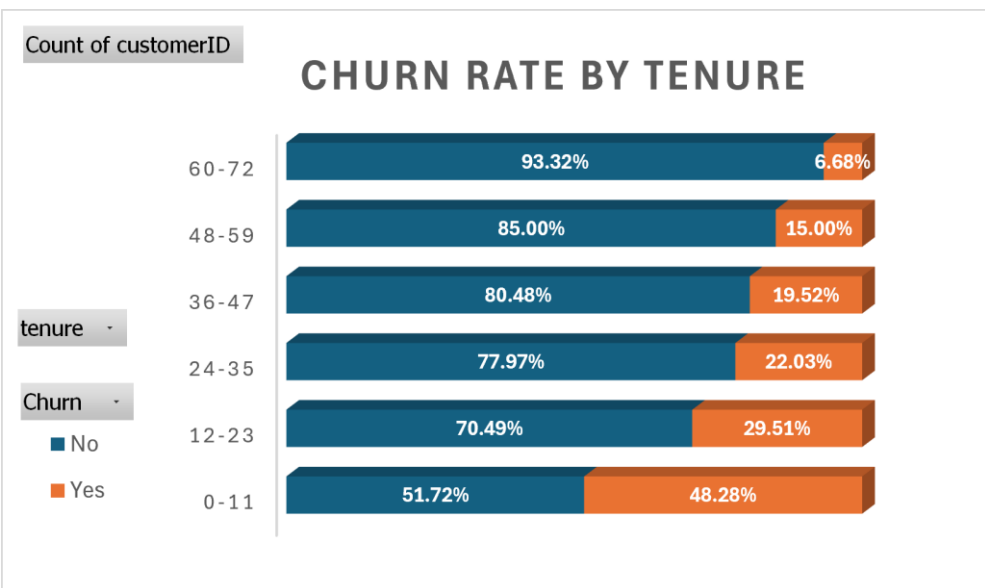


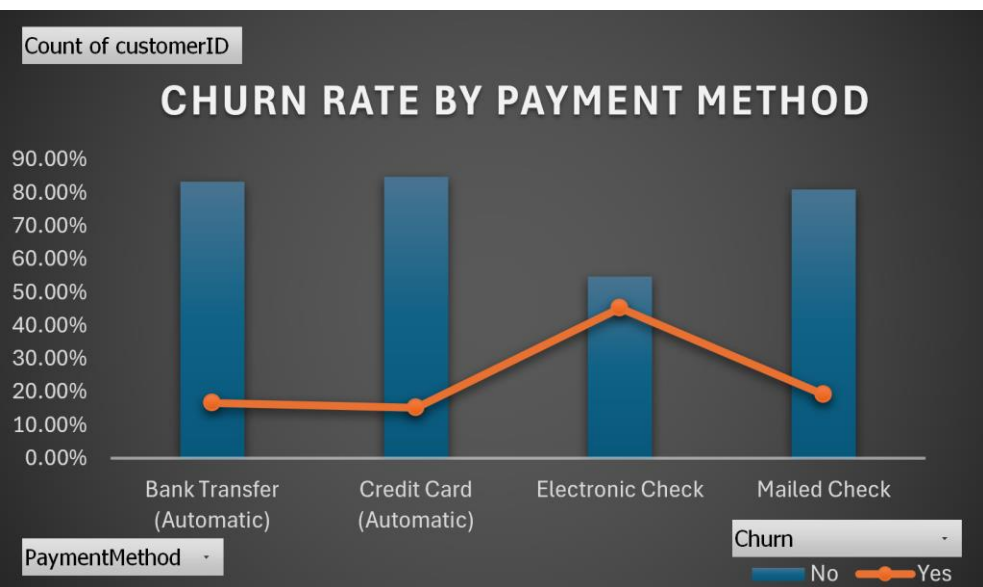
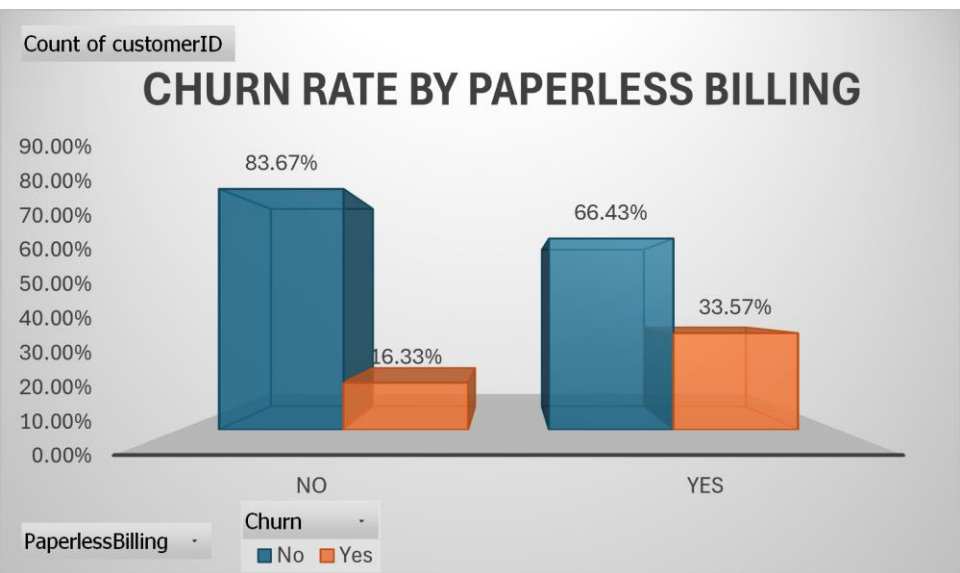
Key Finding

- The customer distribution chart shows that the company has the highest number of customers in the 0–12 months tenure group, followed by customers in the 60–72 months group.
- The churn chart reveals that customers with 0–12 months of tenure have the highest churn rate (48.28%).
- Churn decreases steadily as customer tenure increases, with customers in the 60–72 months group recording the lowest churn rate (6.84%).

Business Interpretation: New customers are significantly more likely to leave the company than long-term customers. As customers remain with the company for longer periods, they become more loyal and less likely to churn. This indicates that customer retention improves with increasing tenure.

Relationship Between the Charts: The customer distribution chart shows that the company has a large number of new customers, while the churn chart shows that this same group experiences the highest churn rate. Conversely, although fewer customers remain in the longer-tenure groups, they have much lower churn rates. Together, these charts indicate that the first year of the customer lifecycle is the most critical period for customer retention, making early engagement and retention strategies essential for reducing overall churn.





Key Finding

- Customers using paperless billing have a significantly higher churn rate (33.57%) than customers who do not use paperless billing (16.33%).
- Among payment methods, customers paying through Electronic Check recorded the highest churn rate (about 45%), while customers using Bank Transfer (Automatic) and Credit Card (Automatic) had the lowest churn rates.
- Customers using Mailed Check also experienced lower churn than those using Electronic Check.

Business Interpretation: Customers who use paperless billing and pay through Electronic Check are more likely to churn than customers using other billing and payment options. This suggests that billing preferences and payment methods are associated with customer retention and may reflect differences in customer experience, convenience, or customer behavior.

Relationship Between the Charts: The paperless billing chart shows that customers enrolled in paperless billing are more likely to churn. The payment method chart further reveals that Electronic Check users experience the highest churn rate, while customers using automatic payment methods have much lower churn. Together, these charts suggest that customers relying on paperless billing combined with Electronic Check payments represent a higher-risk customer segment, whereas customers using automatic payment methods tend to be more loyal and less likely to churn.

CORRELATION HEATMAP

	SeniorCitizen	tenure	MonthlyCharges	TotalCharges
SeniorCitizen	1			
tenure	0.016566878	1		
MonthlyCharges	0.220173339	0.2478999	1	
TotalCharges	0.102410605	0.8258805	0.651064803	1

Key Findings

- Tenure and Total Charges have a strong positive correlation (0.826). Customers who stay longer generally accumulate higher total charges.
- Monthly Charges and Total Charges have a moderately strong positive correlation (0.651), indicating that higher monthly bills contribute to higher lifetime spending.
- Tenure and Monthly Charges have a weak positive correlation (0.248), suggesting monthly charges are only slightly related to how long customers remain.
- Senior Citizen status has very weak correlations with Tenure (0.017), Monthly Charges (0.220), and Total Charges (0.102), indicating age alone is not a strong predictor of these variables.

Business Interpretation: The strongest relationship in the dataset is between tenure and total charges, meaning long-term customers generate significantly more revenue over time. Monthly charges also influence total revenue but are less important than customer longevity. Since Senior Citizen status shows only weak relationships with the financial variables, customer retention strategies should focus more on service quality, pricing, and contract terms rather than age.